Alzheimer Treatment Framework

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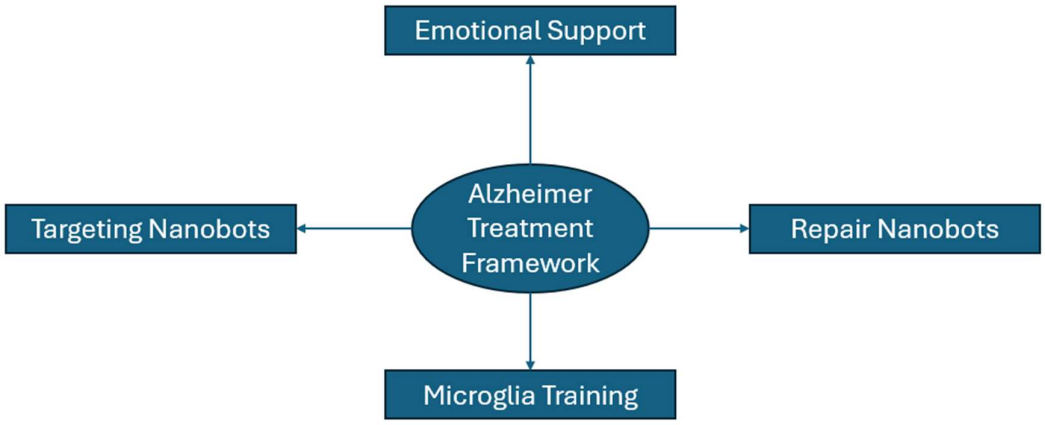
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Introduction

This Alzheimer treatment framework was designed to create a safe and reliable treatment for the neurological disease Alzheimer's. This framework was developed as Alzheimer's is one of the worst neurological diseases that a human being can face. This framework will focus on detection, treatment, repairing and support. This framework is also focused on making sure that the treatment works with the brain not against it.

This framework cannot promise perfect treatment or reliability like any system I have created. However, this framework can promise to provide as much of a reliable/grounded treatment as possible for Alzheimer. This framework was created for neurological researchers and nano engineers to review and potentially implement. Make sure to adapt any design specifications to real-world standards and results of trials.



The diagram above shows the main concepts involved within this framework. We can see that there will be two types of nanobots involved within this treatment plan as well with emotional support for the mental consequences of Alzheimer. We can also see Microglia training which could help build the foundation of immunity.

Alzheimer Detection Section

- **This Alzheimer's detection framework has 5 steps.**
- **The first step is Alzheimer communication**
- **This is where hospitals and medical professionals document and share Alzheimer symptoms online.**
- **This can be done by posting on social media, medical forums or communicating with friends and family.**
- **This step is important because patients need to know what may be wrong with them before visiting a medical professional.**
- **The second step is symptom detection**
- **This is where hospitals and medical professionals check the patient's current symptoms against known Alzheimer symptoms.**
- **This can be done by asking the patient questions, checking prior medical history and doing a check up on the patient.**
- **This step is important to ensure that the patient isn't facing another disease that may appear like Alzheimer.**
- **The third step is Alzheimer verification.**
- **This is where medical professionals check the patient for Alzheimer with more invasive and through examinations and tests.**
- **This can be done through tests such as MMSE, MoCA or through scans such as Amyloid PET scan, Tau PET scan, MRI and CT.**
- **This step is important to ensure that the patient isn't facing other problems and to confirm the presence of Alzheimer.**
- **The fourth step is Alzheimer analysis.**
- **This is where medical professionals analyse the Alzheimer itself figuring out its extent and the stage it is in.**
- **This can be done through using tests like CDR, GDS, FAST and Neuropsychological testing.**
- **This can also further be analysed through repeating the same tests from the verification stage consistently to track the progress.**
- **This step is important to figure out how bad the Alzheimer is and how urgent intervention may be needed.**
- **The fifth step is documentation.**
- **This is where doctors note down their analysis and the data they have on the patient in a structured and clear format**
- **This can be done by using documentation standards such as DSM-5, ICD-11 or NIA-AA criteria.**
- **It is important to remember that documentation can have any format as long as it's understandable to the doctor and their colleagues.**
- **This step is important because it allows doctors to remember key symptoms and act depending on what is happening.**

Alzheimer Nanobot Deployment Section

- **This nanobot deployment framework has 6 steps.**
- **The first step is nanobot training.**
- **This is where medical professionals and doctors train the nanobots to target Alzheimer specific cells.**
- **This can be done by providing the data for the nanobots, running simulations and performing testing with dead cells.**
- **This step is important as the nanobots need to know what they are targeting before they can be deployed within a body.**
- **The second step is nanobot verification.**
- **This is where medical professionals and doctors run diagnostics tests to ensure there are no issues with the nanobots.**
- **This can be done by using diagnostic tools on the nanobots, check connectivity between the nanobots and the external medical devices and checking the nanobot physically for any errors.**
- **This step is important to ensure that the nanobots have no errors before they enter the body.**
- **The third step is nanobot deployment**
- **This is where medical professionals deploy the nanobots into the human brain directly.**
- **This can be done through an Intrathecal which is the safest option for the patient while also allowing the nanobots to get to the brain.**
- **It is important that the targeting nanobots are deployed and used first before the repairing nanobot are deployed.**
- **This step is important because the nanobots need to bypass the BBB while not damaging the body.**
- **The fourth step is Nanobot functions.**
- **This is where the nanobots perform their actions as intended and as expected.**
- **This can be done by letting the targeting nanobots destroy the unhealthy cells and letting the repair bots repair damaged tissue.**
- **This step is important because it is the actual treatment that is used for Alzheimer.**
- **The fifth step is nanobot transmission.**
- **This is where the nanobots transmit the data they have collected from destroying or repairing.**
- **This can be done by the nanobots sending out the signal with their data to the dedicated external medical device.**
- **The sixth step is nanobot disposal**
- **This is where the nanobots dissolve safely into a harmless form that the body can get rid of naturally.**
- **This can be done by giving the commands to dissolve or letting it do so by itself.**

Targeting Nanobot Design

- **I will be discussing 3 areas of this nanobot design which is designed to be safe and reliable to fight against Alzheimer.**
- **The first area I will discuss is size and main material**
- **For size, the nanobot will be 4000 nanometres which is bigger compared to the nanobot created for cancer.**
- **This size was chosen as a bigger battery was needed compared to the cancer treatment nanobot.**
- **For main material, the nanobot will continue to use biodegradable polymer to be able to dissolve safely within the body.**
- **This material was kept as this nanobot is designed to work with the human body not against it.**
- **The second area I will discuss is the physical components.**
- **For the battery, I will be using the solid-state thin-film battery which is the same one I used within the cancer treatment nanobot.**
- **This battery is expected to last about 5-6 hours due its larger capacity compared to the cancer treatment nanobot.**
- **The nanobot will dissolve as an emergency feature at 3% battery for the safety of the patient.**
- **For processing, the nanobot will use an ASIC also known as Application-specific integrated circuit**
- **The ASIC must be configured to allow the nanobot to perform movement, targeting and dissolving.**
- **It is important that for movement the nanobot uses Ameboid type movement to travel around the brain.**
- **For sensors, the nanobot will continue to use Biodegradable electrochemical sensors to locate Tau and amyloid**
- **For storage, the nanobot will use RRAM for the storage of data that will be stored during its deployment**
- **For electrical shielding, the nanobot will use a thin layer of magnesium around itself to protect itself.**
- **This is important because the brain is full of electrical activity that may damage the nanobots if they aren't hardened.**
- **For movement, the nanobot will use silk fibroin for its mechanical legs to allow it to move around even against hard movement.**
- **For targeting, the nanobot will use mechanical arms made from silk fibroin to allow them to extend and then enclose the cell.**
- **It is important that on the surface of the arms, there are Neprilysin (NEP) and Cathepsins which are used to destroy the cells**
- **If the nanobot detects it is dealing with Tau it will release Cathepsins, if it detects amyloid it will release NEP.**
- **The nanobot will detect which one it is using the sensors attached to it.**

Repair Nanobot Design

- **I will be discussing two areas for this repair nanobot design which is designed to repair tissue within the brain.**
- **I will first discuss the size, main material and the systems that are needed**
- **For the size, this nanobot will be 4500 nanometres.**
- **This size was chosen as more capacity is needed to store the collagen as the more collagen there the faster the tissue may heal.**
- **For the main material, the nanobot will mainly use biodegradable polymer and materials.**
- **For the systems, the nanobot will use movement, repair, detection, and dissolving systems to perform its tasks.**
- **This is important because the nanobot just needs to move to the correct areas, detect the damaged areas, repair and dissolve after usage.**
- **For movement, the nanobot must be able to move its mechanical legs effectively to the damaged area.**
- **For detection, the nanobot needs to detect specific markers that damaged area will emit using its sensors.**
- **For repair, the nanobot needs to release all its collagen from its polymer capsule to help with repairing.**
- **For dissolving, the nanobot must dissolve when its battery reaches 3% of if it gets an remote signal giving it to the command to do so.**
- **I will now discuss the physical components.**
- **For battery, the nanobot will use a solid-state thin-film battery.**
- **This battery is expected to last 6-7 hours due to the slow movement it may have.**
- **For processing, the nanobot will use an ASIC to perform all the logic and functions that it needs.**
- **For storage, the nanobot will be using RRAM for the storage of any data that the nanobot needs to store.**
- **For electrical shielding, the nanobot will use a thin layer of magnesium to protect itself from the electrical activity within the brain.**
- **For sensors, the nanobot will use electrochemical sensors to detect the damaged tissue within the brain.**
- **For the storage of collagen, the nanobot will use a polymer capsule to store the collagen until it can be released.**
- **This storage should be between 800 – 1000 nanometres to work effectively.**
- **This is important because the nanobot needs to store the collagen safely before it can be used on the damaged area.**
- **For the mechanical legs, the nanobot will use silk fibroin to ensure that it can move effectively even through more resistant areas.**
- **This is important because the nanobot needs to be able to move around the body easily.**

Microglia Training Section

- **This microglia training framework has 6 steps and is designed to ensure the patient's safety and effective training of the brain's protectors.**
- **The first step is Microglia extraction**
- **This is where medical professionals extract Microglia directly from the patient's brain itself.**
- **This can be done by using post-mortem tissue or surgical samples which both allow for direct extraction from the human body.**
- **This step is important because the microglia are less likely to be rejected if they are recognised and come from the body.**
- **The second step is Amyloid and Tau extraction**
- **This is where medical professionals extract Amyloid or Tau from a patient's brain.**
- **This can be done by using methods such as CSF via lumbar puncture and examining blood plasma.**
- **This step is important because the microglia need actual cells to be able to train against them.**
- **The third step is Microglia training.**
- **This is where medical professionals train the Microglia to target the Amyloid and Tau specifically.**
- **This can be done by using PRRs to mark the extracted amyloid and Tau cells and activate the microglia to attack them.**
- **This process must be repeated and tested consistently to train the Microglia to perform this action correctly.**
- **This step is important because it is where the actual training of the cells is done and it is a relatively new process.**
- **The fourth step is Microglia verification**
- **This is where medical professionals test the Microglia to ensure that it has been trained effectively.**
- **This can be done by testing it against more extracted amyloid and tau cells and comparing results to accurate nanobot targeting results.**
- **The fifth step is microglia reinjection**
- **This is where medical professionals place Microglia back into the body safely.**
- **This can be done by using an Intrathecal which can safely put this back into the brain through the CSF.**
- **The sixth and final step is patient monitoring**
- **This is where medical professionals monitor the patient for any side-effects that reinjection may have caused.**
- **This can be done by using a mixture of vital signs monitor, ECG and blood tests to ensure that the patient isn't experiencing major side-effects.**
- **This step is important to keep the patient safe after reinjection.**

Alzheimer Aftermath Emotional Support Section

- **This emotional support framework has 5 steps, and it is needed as Alzheimer can cause mental health conditions such as depression.**
- **The first step is detection.**
- **This is where medical professionals examine the patient for symptoms that are afflicted with depression or other conditions.**
- **This can be done by talking to the patient, observing their behaviour and talking to any family or support system members.**
- **This step is important because mental health conditions can't be treated if they are unknown.**
- **The second step is identification**
- **This is where medical professionals look at the potential support or actions that need to be taken to support the patient.**
- **This can be done by looking at medical documentation of known treatment options and what can be realistically provided.**
- **This step is important because each mental health condition has its own way it needs to be treated.**
- **The third step is support offer.**
- **This is where medical professionals talk to the patient about support options and if they want to have support.**
- **This can be done by bringing the patient into a private room and presenting the options and ensuring that the patient doesn't feel forced.**
- **If a patient refuses support, they should not be forced to accept it as that can worsen that condition and make them feel uncomfortable.**
- **This step is important because patients need to feel like they are in control of the situation and the support they receive.**
- **The fourth step is recovery.**
- **This is where medical professionals implement the accepted support for the patient.**
- **This can be done by therapy sessions, using healthier coping mechanisms and having a strong support system.**
- **This step is important because the patient must be able to recover from the mental health condition to be fully cured of Alzheimer.**
- **The fifth and final step is monitoring**
- **This is where medical professionals keep an eye on a patient to ensure that there are no further medical issues.**
- **This can be done using appointments, observing patient behaviour and checking up on the patient's support system.**
- **This step is important because some patients can hide conditions as it is conditions like depression that do this.**
- **This emotional support framework can be adapted as needed.**

Alzheimer Patient Treatment Workflow

- **First, the patient sees a post about Alzheimer symptoms online.**
- **Second, the patient realises that they have been struggling with remembering things recently.**
- **Third, the patient decides to book an appointment with their local GP.**
- **Fourth, the GP checks the patient's symptoms against known Alzheimer symptoms.**
- **Fifth, the GP determines that the patient may have brain damage or Alzheimer's.**
- **Sixth, the GP sends the patient to see the neurologist to check for brain damage or Alzheimer.**
- **Seventh, the neurologist performs MMSE, MoCA and CT scans on the patient to check.**
- **If the neurologist identifies that the patient has gotten brain damage, they will provide treatment options for that.**
- **Eighth, the neurologist confirms that patient has Alzheimer due to the results of the scans and tests.**
- **Ninth, the neurologist documents their findings and updates the patient's medical records accordingly.**
- **Tenth, the patient is given an appointment where the nanobots will be deployed into their body.**
- **Eleventh, the targeting nanobots are injected into the patients and perform their job effectively.**
- **Twelfth, the patient is told to come back or an appointment in 2-3 days to allow for the nanobots to dissolve safely.**
- **Thirteenth, the patient is injected with repair nanobots which perform their job effectively.**
- **Fourteenth, the patient is told that another appointment will be booked soon for the Microglia, amyloid and Tau to be extracted**
- **Fifteenth, the extraction was successful and the patient was told that there would be yet another appointment for the rejection of the microglia.**
- **Sixteenth, the patient is reinjected with the microglia with no sudden signs of rejection by their body.**
- **Seventeenth, the patient is offered emotional support if they want it and think that they require it.**
- **If the patient accepts, they get the support. If they reject, the support remains available just unused.**
- **Eighteenth, the patient is monitored for about 2-3 weeks with consistent appointments to check for any signs of side-effects.**
- **Finally, the patient is discharged officially as no signs of Alzheimer or rejection have been detected within the body of the patient.**
- **However, the emotional support for the patient may continue if they want it to.**

Alzheimer Treatment Framework Trade-offs

- **This Alzheimer Treatment Framework has trade-offs just like any other system I have created.**
- **I have identified 5 main trade-offs with this framework.**
- **The first trade-off is time-consumption**
- **This trade-off exists as it will take time to create the nanobots, train the microglia and perform the treatment process overall.**
- **This means that the Alzheimer treatment could take weeks or even months depending on the conditions.**
- **This trade-off can be managed by using automated systems where possible, teams of medical professionals and clear protocols.**
- **The second trade-off is cost.**
- **This trade-off exists as it can and mostly likely will be expensive to develop the nanobots, train staff and perform the cell training.**
- **This means that capabilities may be limited and not everyone can be treated at once or easily.**
- **This trade-off can be managed by using minimalism designs meaning only using what's exactly needed for the nanobots and using existing tools where possible.**
- **The third trade-off is complexity**
- **This trade-off exists as creating the nanobots, training the microglia and perform the whole treatment process can be confusing.**
- **This is because there are multiple verification steps and systems used together to ultimately treat Alzheimer's.**
- **This trade-off can be managed by training staff, allocating teams of professionals and testing the processes to ensure they are understood.**
- **The fourth trade-off is nanobot failure.**
- **This trade-off exists as the nanobots aren't perfect meaning they can fail within the body unexpectedly.**
- **This means that the nanobots will not be able to perform their roles correctly as intended.**
- **This trade-off can be managed by verifying nanobot functionality before deploying, monitoring progress and building in safeguards like the 3% battery safeguard.**
- **The fifth and final trade-off is body rejection.**
- **This trade-off exists as not all patients will accept the reinjection of the microglia easily.**
- **This means that some patients may have immune responses that attack these cells since they aren't recognised within the body.**
- **This trade-off can be managed by monitoring the patients for rejection and ensuring the microglia was extracted from their body.**
- **There may also be more additional trade-offs here as I can't cover all of them.**